ANDREW **CRIX**

TORONTO ON | 416 460 1921 | d.crix@queensu.ca

Education

B.A.Sc.: **Engineering Physics | Final-Year GPA: 4.19/4.3** **06/2026**
Queen's University Kingston,ON

- **Relevant Coursework:** High Performance Computational Physics(Grad Class); Mobile Robotics; Nuclear and Particle Physics(Grad Class); Thesis
- **Scholarships | Awards:** Dean's Scholar; OUA Academic All Star; Most Improved Athlete; Morehead-Cain Scholarship Semifinalist; NSERC USRA; Robert Tyre Jones Jr Scholarship; Harold E. Johns Studentship; Support Our Troops Scholarship; R. H. Clark Applied Science Centennial Scholarship

Academic Exchange: **Physics and Music** **05/2023**
University of St. Andrews St. Andrews, Scotland

Technical Projects

EEG Foundation Model for Heterogeneous Sensor Layouts

- Built a **EEG foundation model in PyTorch** with a CNN-GNN encoder and neuroscience-informed contrastive learning.
- Improved balanced accuracy from **54.9% to 58.9%**; won the **Spotlight Paper Award** for best paper and presentation at the CUCAI.

Generalized Bayesian Estimator for Robotics

- Built a **new nonlinear Bayesian filter** algorithm using finite temperature sampling to **infer posterior uncertainty** from sampled state trajectories.
- Developing a physical differential-drive robot to deploy and validate the algorithm in dynamic environments. Integrated hardware and software.

Automated Recycling Sorter

- Designed mechanical components of an automated recycling sorter, including computer vision (94% test accuracy) for real-time detection and sorting.

Work Experience

Project Associate **05/2026 to Current**
EXP Services Inc **Brampton, Ontario**

- Performed advanced thermal and fluid modelling on structures, completed facade testing, and conducted condition assessments on buildings.
- Identified a reporting bottleneck and built an Agentic AI workflow, reducing engineering report drafting time **from hours to minutes.**

Combat Medic (25 Field Ambulance Reserve Unit) **04/2025 to Current**
Canadian Armed Forces (Royal Canadian Medical Service) **Toronto, Ontario**

- Provided medical support to 32 Canadian Brigade Group, administering emergency medical care in **high-pressure environments**.
- Trained as a sentinel, providing informal support to colleagues struggling with personal or professional challenges.

Founder/Private Piano Teacher **05/2020 to Current**
Self-Employed **Toronto, ON | Remote, ON**

- Teach personalized piano lessons. Built a **Django-based piano learning platform with 200+ lessons**, integrating CI/CD deployment workflows.
- Developed a **custom audio ML architecture integrating pretrained audio transformers** for piano pitch recognition and classification.

Research Assistant (NSERC USRA) **05/2024 to 08/2024**
Queen's University - Dr Jackson Crane **Kingston, ON**

- I developed a detonation visualization system with a custom high-speed LED pulser PCB, adopted as Crane Energy Group's **primary diagnostic**.
- I synchronized ignition, imaging, and pressure sensing to meet precise timing. I analyzed the data using Python to extract detonation characteristics.

Software Engineer Intern **05/2022 to 08/2022**
Cancer Care Ontario - Sunnybrook Hospital **Toronto, Ontario**

- I built and deployed a communication dissemination system over a relational schema with role-based access, used by Sunnybrook after my departure.
- Created administrative workflows and automated tracking features, enhancing reliability, visibility, and usability of the system.

Volunteer/Extracurriculars

Queen's University

- **Varsity rower** and mentor for junior athletes, co-chaired the Queen's Space Conference, led orientation programming for 1,000+ students, QMIND design team member, participated in engineering design competitions.

Technical Skills

Programming: Python, C++, MATLAB/Simulink, SQL, React, CSS
Libraries/Tools: PyTorch, TensorFlow, Linux, Git, NumPy, SciPy, pandas, mpi4py, scikit-learn, OpenCV, PyTorch Geometric, RESTful API